

Dynamics of the West African Monsoon Jump

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ABSTRACT

The observed abrupt latitudinal shift of maximum precipitation from the Guinean coast into the Sahel region in June, known as the West African monsoon jump, is studied using a regional climate model. Moisture, momentum, and energy budget analyses are used to better understand the physical processes that lead to the jump. Because of the distribution of albedo and surface moisture, a sensible heating maximum is in place over the Sahel region throughout the spring. In early May, this sensible heating drives a shallow meridional circulation and moisture convergence at the latitude of the sensible heating maximum, and this moisture is transported upward into the lower free troposphere where it diverges. During the second half of May, the supply of moisture from the boundary layer exceeds the divergence, resulting in a net supply of moisture and condensational heating into the lower troposphere. The resulting pressure gradient introduces an inertial instability, which abruptly shifts the midtropospheric meridional wind convergence maximum from the coast into the continental interior at the end of May. This in turn introduces a net total moisture convergence, net upward moisture flux and condensation in the upper troposphere, and an enhancement of precipitation in the continental interior through June. Because of the shift of the meridional convergence into the continent, condensation and precipitation along the coast gradually decline. The West African monsoon jump is an example of multiscale interaction in the climate system, in which an intraseasonal-scale event is triggered by the smooth seasonal evolution of SSTs and the solar forcing in the presence of land–sea contrast.

1. Introduction

The West African monsoon system is a climatological feature of major economic and social importance to the population of the region whose economy heavily relies on agriculture. Understanding its dynamics, variability at various time scales, and ultimately improving our skill in predicting its onset and evolution would contribute toward food security and the stability of the region. For example, information on the timing of the onset of the monsoon, that is, the first rains that are sufficient to provide enough soil moisture at the time of planting and uninterrupted by prolonged dry conditions, would enhance agricultural productivity.

The monsoon season begins with intense rainfall near the Gulf of Guinea in April, which typically remains in place until the end of June. A secondary precipitation maximum develops near 10°N in late May. During the last week of June, on average, the coastal precipitation

maximum shifts to the latitude of the secondary maximum over a few days. For the rest of the rainy season, the precipitation maximum follows the seasonal cycle fairly accurately. So the monsoon “jump” can be characterized as the onset of intense convection and rainfall along 10°N accompanied by its sudden termination along the Guinean coast. This is in sharp contrast with the fact that the ultimate cause of the seasonal migration of the ITCZ, the northward march of the sun, follows a smooth and continuous cycle.

Currently, there are contrasts between observational studies of the monsoon jump and the theory of monsoon circulations. The purpose of this paper is to improve our understanding of the atmospheric hydrodynamics of the jump by analyzing the fundamental physics of the jump in an atmospheric model that reproduces its observed characteristics. We do not simulate a particular year, but adopt a “climate mode” approach in which a representative monsoon onset season is generated in the model with climatological surface and lateral boundary forcing. While recent improvements in seasonal forecast skill over West Africa are encouraging, improved insights about the basic dy-

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namics of the jump will provide more relevant predictors of its timing, as well as the overall intensity and distribution of rainfall during the monsoon season.

Our current state of knowledge of monsoon jump processes over West Africa is reviewed in the following section. A description of the regional climate model is provided in section 3, and the simulated monsoon jump is compared with the observed. In section 4, a systematic analysis of the moisture, momentum, and energy conservation laws is performed to isolate the physical processes responsible for the monsoon jump in the model, and a chronological summary of the process is provided. In the last section, the results are discussed in the context of previous studies on the subject and predictors of the jump are specified.

2. Background

Sultan and Janicot (2000) performed an observational analysis of the discontinuity in the northward migration of the ITCZ using gridded station rainfall data compiled at the Institut de Recherche pour le Développement (IRD), the Agence pour la Sécurité de La Navigation Aérienne en Afrique et à Madagascar (ASECNA), and the Comité Interafricain d'Études Hydrologiques (CIEH) as well as the National Centers for Environmental Prediction–National Center for Atmospheric Research (NCEP–NCAR) reanalysis (Kalnay et al. 1996) for 1968 to 1999. They suggested that the jump is an acceleration of the seasonal cycle triggered by westward-propagating intraseasonal-scale atmospheric circulations. This phenomenon is also documented in an observational study by Le Barbe et al. (2002), who showed that 5° and 10°N are preferred locations for precipitation maxima. Thus, the classic vision of a continuous seasonal migration of the ITCZ giving rise to two rainy seasons along the Coast of Guinea and a single one over the Sahel was called into question.

Using the same rain gauge data as Le Barbe et al. (2002), along with Meteosat IR images, the Estimation des Précipitations par Satellite (EPSAT)-Niger network, and the NCEP–NCAR reanalysis, Lebel et al. (2003) further confirmed the discontinuity in the northward migration of rainfall over West Africa. They identified two distinct precipitation regimes. The first, referred to as the oceanic regime, involves the seasonal shift of rain onto the coast from the tropical Atlantic. The second phase, the continental regime, consists of a sudden rise in mean daily rainfall and the number of rain events near 10°N. The continental regime typically starts at the end of June and accounts for 75% to 90% of the total precipitation at this latitude. They also

showed that interannual variability of Sahelian rainfall is associated with fluctuations in the number of rain-efficient convective systems within the continental regime.

In a detailed study of the monsoon onset in the observations, Sultan and Janicot (2003) suggest that the jump involves complex interactions among convective processes in the ITCZ, African easterly jet dynamics, and circulations associated with local topography. They propose that the Ahaggar Plateau and the Tibesti highlands could exert significant control on the atmospheric circulation and convection in the ITCZ. Their explanation of the process can be summarized as follows: About 40 days before true onset, the ITCZ is located near 5°N and the intertropical front (ITF; the line of confluence between the southwesterly moist flow and the northeasterly dry flow) is located near 15°N. Dynamics associated with the heat low dominates the continental region. During this time, isolated convective systems appear over the continent, resulting in an apparent expansion of the ITCZ. About 10 days before the jump, the heat low is enhanced by the interaction of the northeasterly winds with orography. This occurs when the southward advection of dry air cause temporary rainfall decreases south of the ITF and increased radiative heating of the surface. Meanwhile, the seasonal development of the low-level westerly flow and the midtropospheric African easterly jet increases the vertical wind shear, which increases the local potential instability. The heat low eventually introduces a reversal in the potential vorticity gradient as well as the generation of African easterly waves and convection. The release of potential instability and subsequent inertial instability then shift the ITCZ to about 10°N.

A satellite-based study by Gu and Adler (2004) uses rainfall data from the Tropical Rainfall Measuring Mission (TRMM) and surface winds from the Quick Scatterometer (QuikSCAT), along with the NCEP–NCAR reanalysis, to show that rainfall tends to be concentrated along two latitudes, 5° and 10°N. They found that the precipitation field lies mainly along 5°N during April, May, and June and it is mainly concentrated along 10°N during July, August, and September. They suggested that the appearance of intense rainfall along the Guinean coast in April is related to the occurrence of warm SSTs over the tropical eastern Atlantic, and its disappearance might be related to the formation of an oceanic cold tongue complex over the region. They conclude that the appearance of the rainfall along 10°N during the end of June is independent of the Gulf of Guinea precipitation system and the time of the onset of rainfall events at 10°N coincides with the northward shift of the African easterly jet and its associated ver-

tical and horizontal shear zones and the low-level westerly flow, along with the development of westward-propagating African easterly waves. Okumura and Xie (2004) used the Center for Climate System Research (CCSR)/National Institute for Environmental Studies (NIES) GCM and showed that the interaction of the monsoon with the equatorial cold tongue might be responsible for the accelerated northward shift of the rainband. According to their result, the monsoon circulation also accelerates the cooling process in the equatorial Atlantic due to surface southerlies, which induce local upwelling.

Most recently, Ramel et al. (2006) and Sijikumar et al. (2006) used a regional numerical model to simulate the monsoon jump of 1992 and suggested that it was related to an abrupt northward shift of the heat low from 10°–15°N to 20°–25°N. They also showed the existence of some correlation between the location of the easterly flow and the location of rainfall maximum. Similar work by Sijikumar et al. (2006) suggested that the onset of the monsoon is characterized by an increase in the monsoon flow due to a deepened heat low.

Many previous theoretical models of monsoon onset have been presented in the context of zonal symmetry in a moist, geostrophic atmosphere (e.g., Plumb and Hou 1992; Eltahir and Gong 1996). According to these models, monsoon onset occurs because a strictly zonal flow in thermal equilibrium is unsustainable when the absolute vorticity vanishes somewhere inside the system. These ideas are based on the concept of inertial instability, which can occur when a parcel of air in a geostrophic zonal basic state, u_g , is displaced by a distance, δy . The meridional acceleration of the parcel is governed by

$$\frac{Dv}{Dt} = -f \left(f - \frac{\partial u_g}{\partial y} \right) \delta y, \quad (1)$$

where f is the Coriolis parameter and the term in parentheses is the absolute vorticity. According to Eq. (1), when a parcel of air is displaced northward ($\delta y > 0$) in the Northern Hemisphere ($f > 0$) into a region of negative absolute vorticity, it is accelerated further northward. Therefore, a change in sign of absolute vorticity is considered to be the condition for inertial instability, which introduces meridional flow in an otherwise zonal geostrophic flow. These models relate the change in absolute vorticity to the thermal wind, meridional pressure gradients, and ultimately, the meridional boundary layer entropy gradient (Emanuel 1995).

The onset of the West African monsoon may be more complicated than these models suggest. Because of the complexity of the background flow over West Africa in summer, advective processes, deviations from

geostrophy, and prominent wave activity and preonset shallow meridional circulations are likely to play important roles. Nevertheless, these models have contributed to the improvement of seasonal prediction over West Africa. The primary conclusion of these studies is that the latitudinal gradient in heat content is an important determinant of the timing of the onset and strength of the monsoon. Fontaine et al. (1999) used the NCEP–NCAR reanalysis to demonstrate that using the meridional moist static energy (MSE) gradient between the coast of Guinea and the Sahara, along with oceanic indicators, significantly improves prediction skill. Similarly, by performing statistical analysis on Climatic Research Unit (CRU) rainfall data along with the NCEP–NCAR reanalysis, Fontaine and Philippon (2000) found that the seasonal evolution of moist static energy over West Africa is particularly relevant to the intensity of the monsoon, with a stronger monsoon signal preceded by a large northward translation of the MSE maximum.

3. Simulation

a. Model description and simulation design

The tropical regional climate model (RCM) used in this study is an adaptation of the fifth-generation Pennsylvania State University–NCAR Mesoscale Model (MM5, version 3; Grell et al. 1994), modified as in Vizi and Cook (2002). The model is nonhydrostatic and it solves the equations governing horizontal and vertical momentum, temperature, pressure, moisture, and liquid water equations on σ surfaces. The full three-dimensional Coriolis force is included.

The model simulation is run over a rectangular domain enclosed by 30°S to 45°N and 50°E to 50°W. The grid spacing is 90 km. There are 23 vertical levels and the model time step is 90 s. The top of the atmosphere is fixed to be at 50 hPa for this tropical application and an upper radiative boundary condition is used. The effects of snow cover are neglected.

The model is initialized with climatological conditions from the NCEP–NCAR reanalysis and integrated from 15 April to 15 August, and the first 15 days of output are discarded as the spinup period. The NCEP–NCAR reanalysis is also used for lateral boundary conditions, with monthly means from the reanalysis taken to represent the middle of the month and a linear interpolation in time used to generate the boundary conditions for every 12 h. SSTs are also prescribed using the reanalysis and interpolated as described above to generate 12 hourly values from the monthly climatological values. Surface albedo, moisture availability, emissivity, and thermal inertia of the surface are deter-

mined from the model's lookup table based on U.S. Geological Survey (USGS) land use categories (Grell et al. 1994).

The Kain–Frisch scheme (Kain and Fritsch 1990) is used to parameterize convection. The closure assumes that the convective available potential energy is almost entirely removed at every time step at which convection occurs. This scheme incorporates the effects of entrainment and detrainment as well as downdrafts associated with evaporation of rain on the convection and large-scale environment and it produces realistic precipitation distribution (Vizy and Cook 2002).

Ground temperature is calculated from the surface heat budget using a force–restore method developed by Blackadar (Zhang and Anthes 1982). The heat flow into the ground is calculated by using the bulk aerodynamic formula. Diffusive and latent heat fluxes are calculated using the Blackadar high-resolution planetary boundary layer (PBL) scheme (Blackadar 1979), which uses the Monin–Obukhov similarity model. The nature of the similarity functions used depends on the stability (i.e., the Richardson number) within the PBL. Horizontal diffusion in the model is parameterized in such a way that it controls aliasing and nonlinear instability. Second-order diffusion is used along the boundaries and fourth-order is used elsewhere.

The radiation scheme is adapted from that of the NCAR Community Climate Model 2. Its shortwave scattering/absorption is parameterized by the δ -Eddington approximation (Joseph et al. 1976) and is applied over 18 spectral intervals. The short-wave optical properties of the clouds depend on droplet size and the liquid water path. The scheme treats longwave absorption by ozone and CO₂ using the broadband absorption technique of Kiehl and Briegleb (1991). Longwave broadband emissivity of clouds is a negative exponential function of liquid water path.

b. Observations of the monsoon jump, and comparison with the model simulation

Figure 1a displays the time series of precipitation from the RCM simulation on the model's 90-km grid. The output is averaged between 10°E and 10°W where the coastline is parallel to the equator and land surface features such as topography and land use type have zonal uniformity. The model's 3-hourly output is averaged to form daily means. The vertical line marks the Guinean coastline, and the dashed line denotes the approximate date of the monsoon jump, that is, when the precipitation maximum becomes established in the region around 10°N (hereafter referred to as the continental interior). An intense precipitation band develops near 4°N in early May and its latitudinal extent

shrinks through June. Meanwhile, precipitation develops in the continental interior and gradually increases in intensity. The criteria used by Sultan and Janicot (2003) for the choice of the date of the jump is generalized here to be applicable to modeled precipitation as well as other fields. Here the jump of a maximum of precipitation or any other field from the coast in to the continent is said to have occurred if the 10-day average of the field along the 10°N becomes greater than that at near 5°N. The continental precipitation becomes greater than the coastal (oceanic) precipitation on about 27 June, defined as the date of the monsoon jump in this simulation. After the jump, the continental precipitation migrates northward following the solar forcing. The coastal precipitation dies away after the jump, falling below 2 mm day^{−1} after the first week of July.

Similar trends are displayed in the observed climatology but there are important differences from the simulation. Figures 1b and 1c display the daily mean precipitation from the TRMM (Nicholson et al. 2003) and the Global Precipitation Climatology Project (GPCP; Huffman et al. 2001), respectively. Both are averaged over 1998–2004. The horizontal resolution of both of these observations is 1° of latitude and longitude, only slightly coarser than that of the RCM output so no interpolation has been done for the comparison. In the GPCP observations the jump occurs around the 3 July, and in the TRMM data it occurs about one week later, on 10 July. Note that even though these observations are averaged over the same time period, the date of the jump is not the same. Unlike in the RCM simulation, precipitation rates greater than 3 mm day^{−1} persist until the end of July (not shown) along the coast. Similar to the simulation, the continental precipitation is in place as early as the middle of May, but with magnitudes lower than those along the coast.

A longer-term representation of the precipitation climatology is available from the Climate Prediction Center (CPC) Merged Analysis of Precipitation product (CMAP; Xie and Arkin 1997) shown in Fig. 1d. These are available only for 5-day means, and the horizontal resolution is only 2.5° of latitude and longitude, so the jump is not well resolved. However, many of the features of the precipitation development seen in the model and higher-resolution data are in evidence. Relatively weak continental precipitation develops by mid-May and, in June, the latitudinal extent of the oceanic precipitation shrinks. The jump occurs on roughly 14 July, more than 2 weeks later than in the RCM, 3 days later than in the TRMM data, and 10 days later than in the GPCP observations for 1998–2004. After the jump, precipitation rates up to 5 mm day^{−1} persist through August along the Guinean coast.

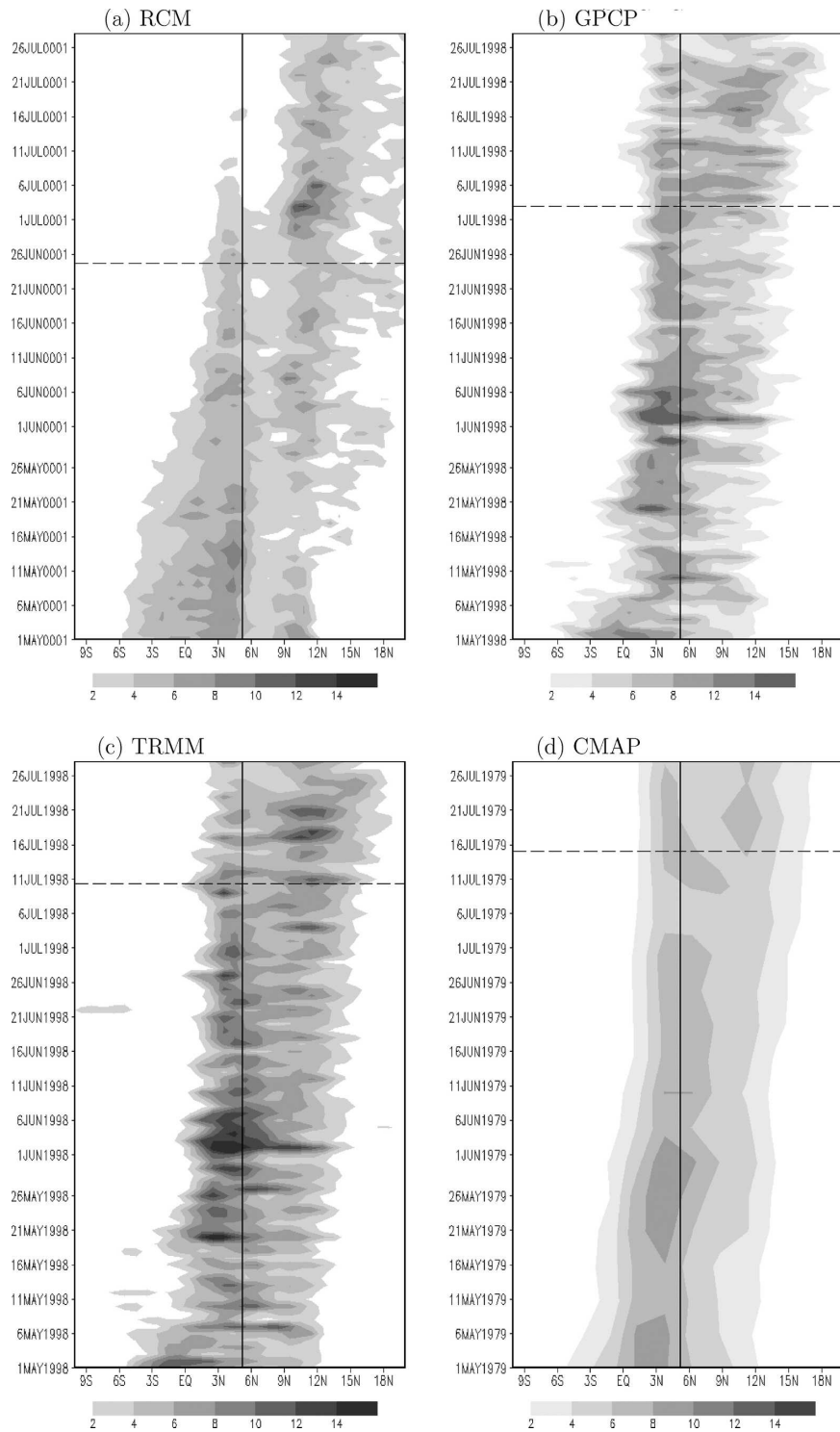


FIG. 1. Precipitation from the (a) RCM simulation, (b) GPCP 6-yr mean, (c) TRMM 6-yr mean, and (d) CMAP 6-yr mean. All fields are averaged between 10°E and 10°W and are in mm day⁻¹. Dashed lines indicate approximate date of the monsoon jump, and the bold lines represent the approximate latitude of the coastline.

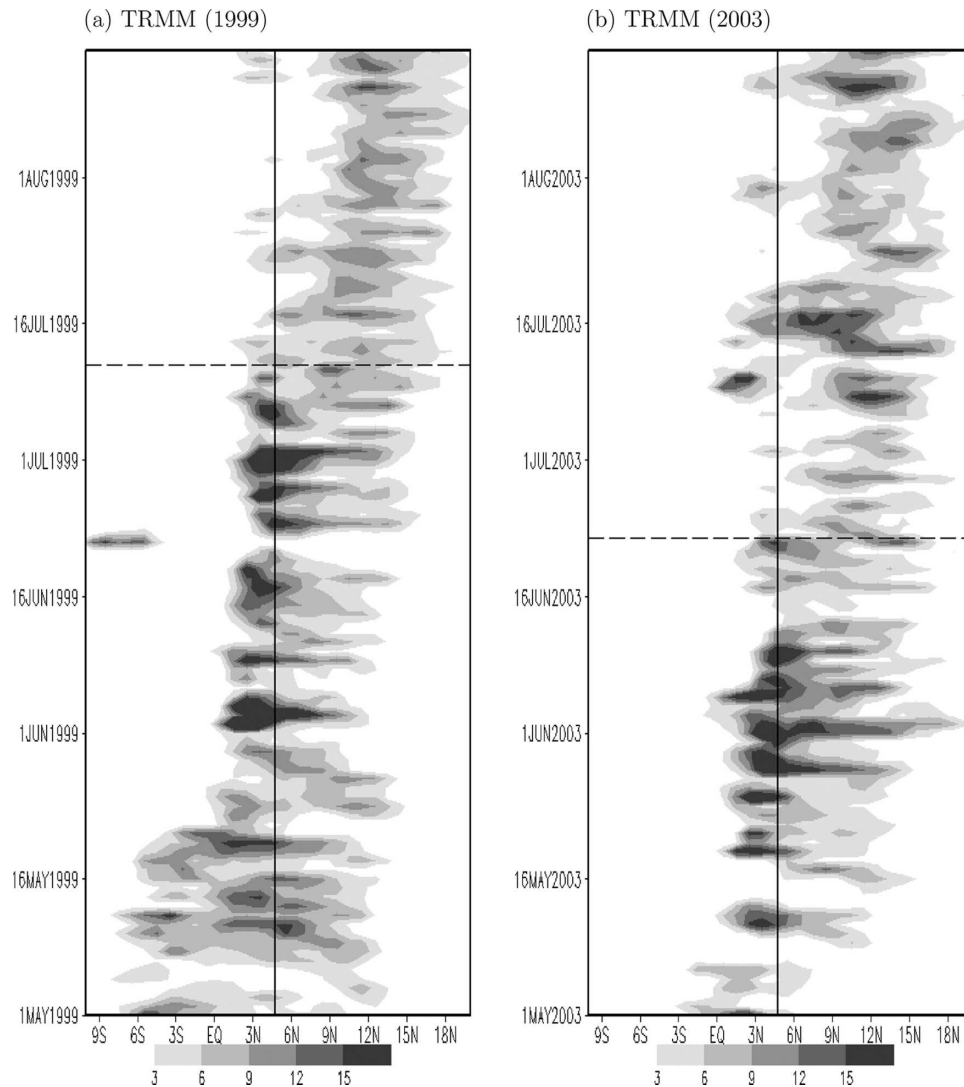


FIG. 2. Precipitation from TRMM (3B42) for the years (a) 1999 and (b) 2003. Both fields are averaged between 10°E and 10°W and are in mm day^{-1} . Dashed lines indicate approximate date of the monsoon jump, and the bold lines represent the approximate latitude of the coastline.

Since averaging over even a few years may obscure the sharpness of the monsoon jump, and the RCM simulation is a single realization of climate and not an average, we also present observations for individual years. Figures 2a and 2b illustrate the evolution of West African precipitation during 1999 and 2003, respectively, from the TRMM data. Overall, the precipitation field for 1999 is more similar to the climatologies (Figs. 1b, c, d) than 2003. Significant rainfall develops over the continent during May 1999, but it is weaker during 2003. Progressive shrinking of the oceanic precipitation during June occurs in 1999, but in 2003 it is less pronounced. The prejump continental precipitation during June 2003 is stronger than that of 1999 and, as shown by

the analysis below, this may be partly responsible for the earlier jump in 2003 (near 23 June) when compared to that of 1999 (near 10 July). This degree of interannual variability is also registered in the observational analyses (Sultan and Janicot 2003; Fontaine and Louvet 2006). They reported that the standard deviation of about 7–8 days when filtered series are used and when the series are unfiltered the standard deviation was found to be about 15 days. Under a slightly different experimental setup, Sijikumar et al. (2006) also verified MM5's ability to simulate the West African monsoon onset.

The monsoon jump is also manifested in the National Oceanic and Atmospheric Administration (NOAA)

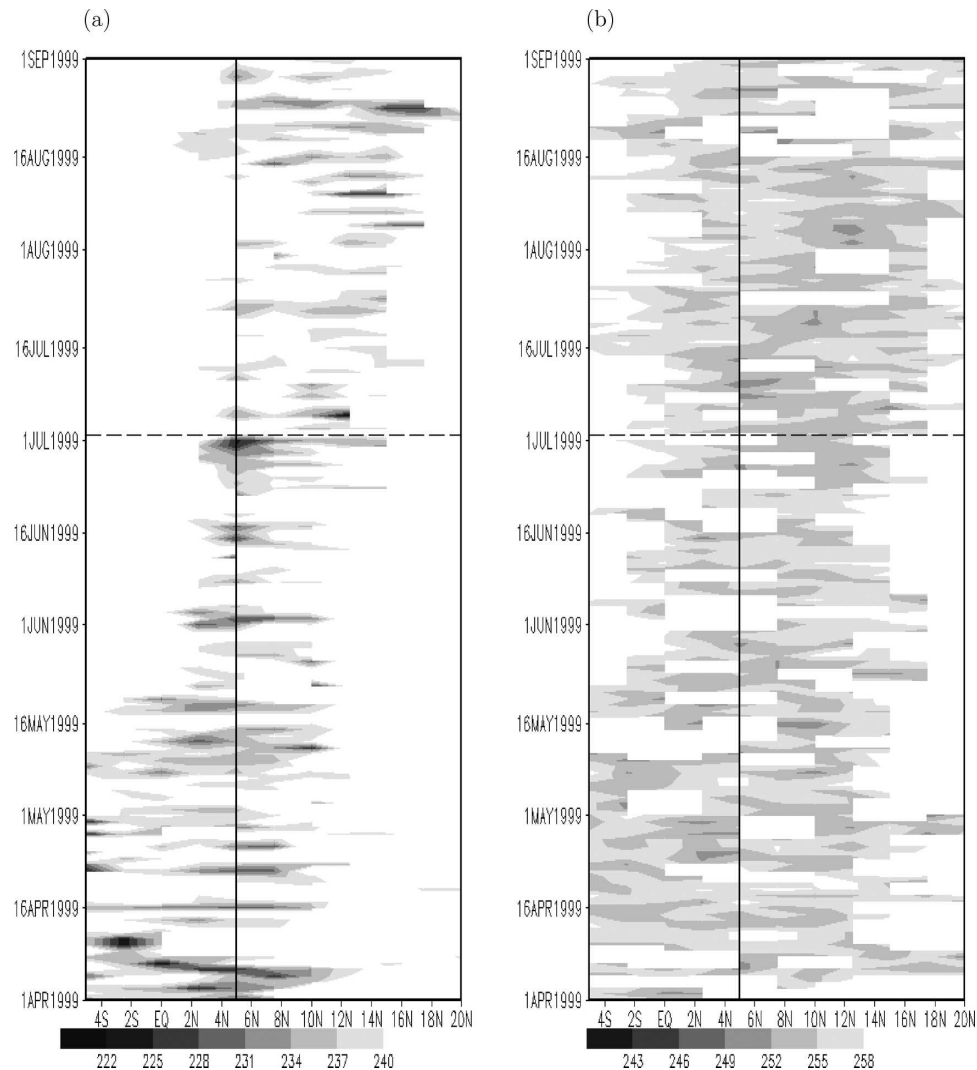


FIG. 3. Blackbody cloud temperature: (a) 220–240 and (b) 243–258 K. Calculated from NOAA CDC outgoing longwave radiation data. Averaged between 10°E and 10°W. The bold line represents the approximate latitude of the coastline. The dashed line represents the approximate date of the shift of the temperature minimum into the continent.

daily outgoing longwave radiation (OLR) data (Liebmann and Smith 1996), converted to cloud-top temperature using the Stefan–Boltzmann relation. Figure 3a shows a time series for 1999, averaged between 10°E and 10°W, of the location of clouds with blackbody temperatures lower than 240 K. This corresponds to pressure levels of approximately 300 hPa and above. Many of the features of the monsoon jump identified in the precipitation data and the model are reflected in the evolution of the high cloud distribution, including the narrowing of their latitude width from May to the end of June and the subsequent jump into the continental interior.

The OLR signal for blackbody temperatures of

240–260 K is shown in Fig. 3b. This temperature range approximates the locations of lower clouds, while roughly filtering out emission from the surface and near-surface water vapor. There are indications of significant amounts of low clouds in the continental interior throughout the spring, but the jump is not apparent.

The monsoon jump occurs in the unique large-scale dynamics of the West African summer, and the RCM is able to capture that environment accurately. For example, May through July 925-hPa winds and geopotential heights from the model are compared with the NCEP–NCAR reanalysis (Kalnay et al. 1996) in Fig. 4. Of particular relevance for the physics of the monsoon jump (discussed below) is the evolution of the conti-

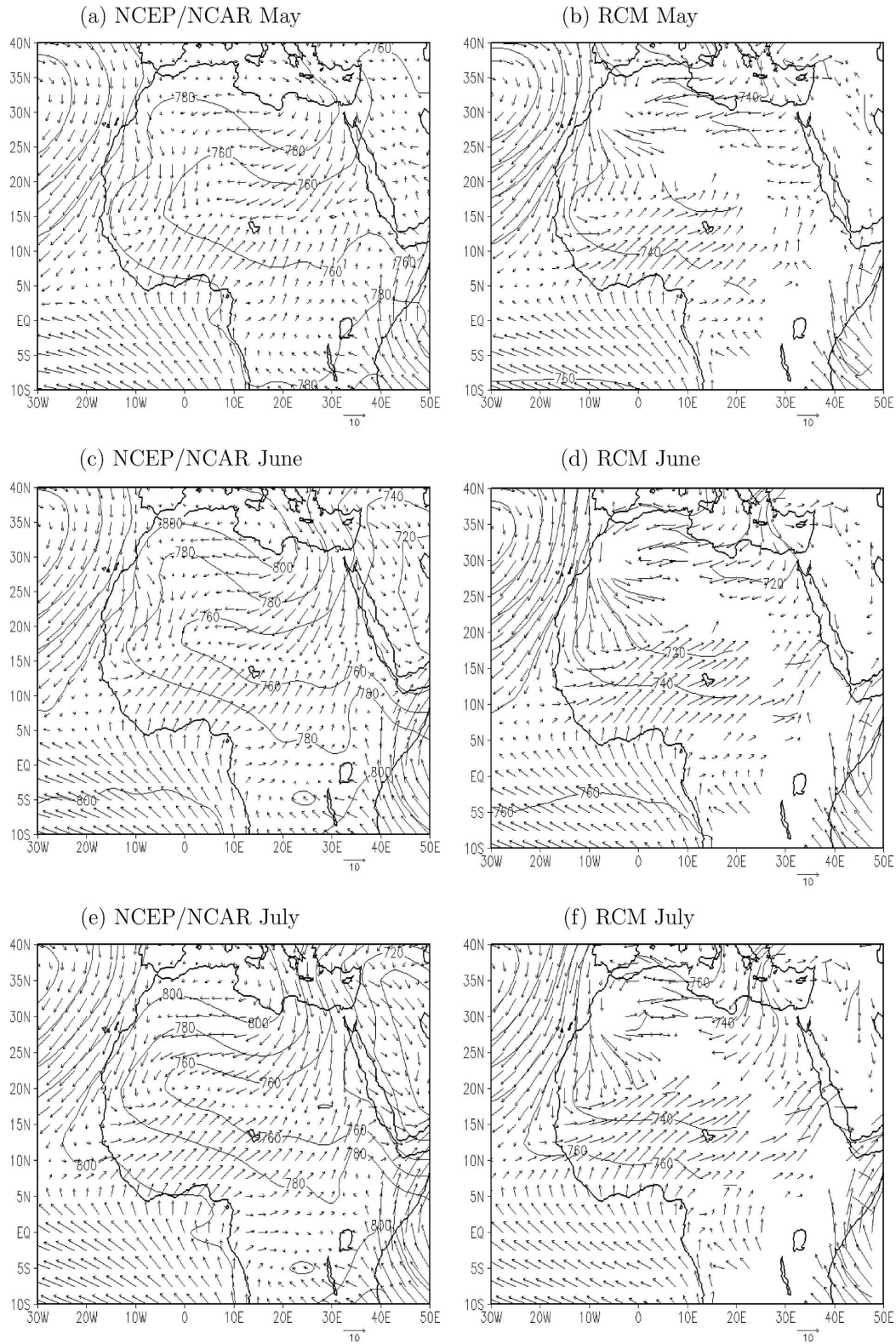


FIG. 4. Monthly mean wind (m s^{-1}) and geopotential height (gpm) at 925 hPa from NCEP–NCAR reanalysis and RCM simulation.

mental low pressure system and low-level flow. Comparison of the distance between the 780- and 760-m geopotential height contours in the reanalysis shows that this low pressure system intensifies as the season progresses. A similar increase in pressure gradients is also evident in the RCM (e.g., the distance between the 760- and 740-m contours). As the low strengthens, the line of confluence between the southwesterly monsoon flow and the northeasterly Harmattan flow moves from about 15°N in May to about 20°N in July in both the reanalysis and the model simulation. The strong low-level southerly flow into the continent is in place during May and shows little variation as the season progresses. Further validation is also provided by Vizu and Cook (2002).

4. Analysis

The vertical structure of the condensation field is considered because the OLR data presented in Fig. 3 suggest that the monsoon jump involves an abrupt shift in upper-tropospheric clouds. The free troposphere in the model can be classified into two layers in which the condensation field evolves in a distinct manner. Figures 5a and 5b show the development of the condensation field in the upper layer (above 525 hPa) and the middle layer (825–525 hPa), respectively.

Many features of the precipitation field (Fig. 1a) are reflected in the upper-layer condensation field (Fig. 5a). The oceanic condensation field narrows throughout the prejump period and there is a marked discontinuity in its evolution near the time of the precipitation jump on 25 June. Significant amounts of upper-layer condensation (and precipitation) are present over the continent from the end of May through June. After the jump, the upper-level condensation follows the seasonal cycle deeper into the continent as in the precipitation and OLR observations.

The seasonal evolution of the middle-layer condensation is significantly different. It is established near 10°N as early as the middle of May, and intensifies in the end of June. Its conspicuous absence along the coast and its early appearance suggest that it is mainly related to land surface processes.

The moisture, momentum, and energy budgets of the system are analyzed to understand the relationship between processes in the upper and middle layers and their interactions with the boundary layer and the surface. A chronology of the entire process is given afterward.

a. Monsoon jump processes

To isolate the physical processes responsible for the evolution of the upper-layer condensation field, the

moisture budget equation is integrated from the bottom of the layer at $p_m = 525$ hPa to $p_t = 50$ hPa, given by

$$\int_{p_m}^{p_t} (\dot{q}) dp = \int_{p_m}^{p_t} \left(\frac{\partial q}{\partial t} \right) dp + \int_{p_m}^{p_t} \left(\frac{\partial uq}{\partial x} + \frac{\partial vq}{\partial y} \right) dp - \omega q_{p_t} + \omega q_{p_m} + \int_{p_m}^{p_t} F_q dp. \quad (2)$$

The left-hand side of Eq. (2) is the net change in mixing ratio by evaporation and condensation (if negative) and the right-hand side includes the local time rate of change, the zonal and meridional components of the moisture convergence, and the vertical fluxes; F_q is the sum of horizontal diffusion, vertical diffusion, and the nonhydrostatic divergence of moisture.

Figures 6a and 6b display the vertical flux of moisture from the middle layer, $-\omega q_{p_m}$, and the horizontal convergence of moisture within the upper layer, respectively. These are the two largest contributors to the upper-level condensation. The sum of the other terms in Eq. (2) accounts for less than 10% of the total condensation. As marked by the dotted line in Fig. 6a, the vertical moisture flux into the upper level over the continent (near 10°N) begins on 2 June, nearly 4 weeks before the monsoon jump (dashed line) in the simulation. From this time till the occurrence of the jump, two regions of vertical flux are maintained, reminiscent of a “double ITCZ” structure. After the jump, the vertical moisture flux along the Guinean coast falls below 2 mm day⁻¹ within a week.

Between 2 and 25 June, a small horizontal moisture convergence helps to maintain the upper-level condensation along the Guinean coast (Fig. 6b), and this term is also important during the second half of May in the continental interior.

The next step is, then, to identify the sources of moisture flux from the middle to the upper layer. The moisture budget for the middle-layer can be written as

$$-\omega q_{p_m} = - \int_{p_b}^{p_m} \left(\frac{\partial uq}{\partial x} \right) dp - \int_{p_b}^{p_m} \left(\frac{\partial vq}{\partial y} \right) dp - \omega q_{p_b} + \int_{p_b}^{p_m} (\dot{q}) dp - \int_{p_b}^{p_m} \left(\frac{\partial q}{\partial t} \right) dp - \int_{p_b}^{p_m} F_q dp. \quad (3)$$

The left-hand side, $-\omega q_{p_m}$, is the moisture flux between the middle and the upper layer shown in Fig. 6a, and the variables are defined as in Eq. (2). Figure 7 displays the first four terms from the rhs of Eq. (3). The last two terms are small. The shadings are changed relative to Eq. (2) for clarity, so that the sources of moisture in the middle level that supply the flux into the upper layer

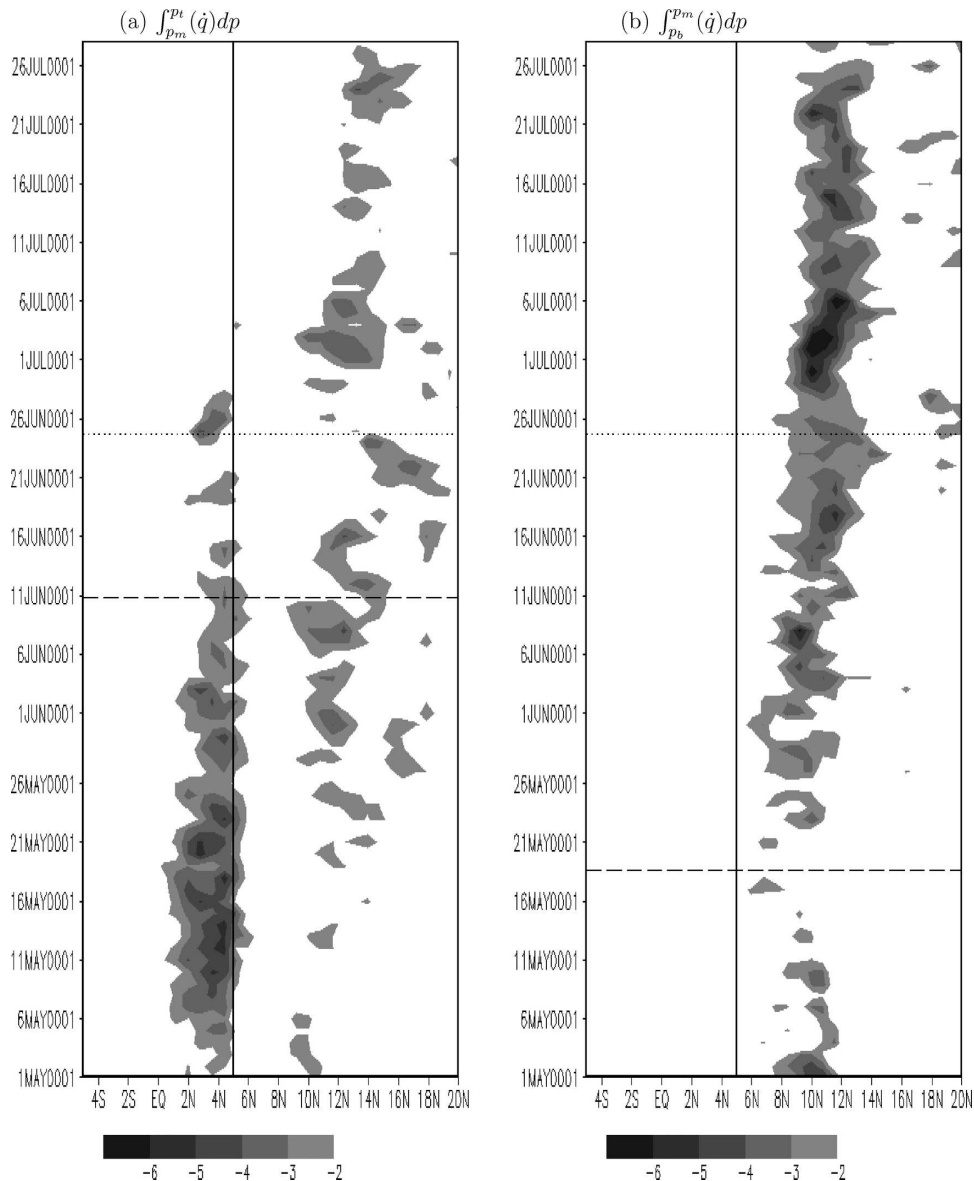


FIG. 5. Vertical integral of moisture condensation (a) between 525 and 50 hPa and (b) between 825 and 525 hPa in the model in mm day^{-1} ; both are averaged between 10°E and 10°W . The bold line represents the approximate latitude of the coastline, and the dashed lines represent the approximate dates of onset and the start of preonset precipitation (see text).

are positive (darker shading) and the sinks are negative.

The two main sources of moisture in the middle layer are meridional moisture convergence (Fig. 7a) and the vertical flux from the boundary layer below (Fig. 7b). The abrupt jump in deep convection and upper-layer condensation around 2 June (Fig. 6a) coincides with a similar jump in the middle-layer meridional moisture convergence (dashed line; Fig. 7a), but the vertical flux from the boundary layer is relatively constant throughout the season. This vertical flux is a source of moisture

for the middle layer in the continental interior throughout the season.

Figure 7c shows that the zonal divergence of moisture significantly reduces middle-layer moisture levels almost everywhere north of 4°N , and is strong enough to compensate the meridional convergence of moisture at times. For example, in May, zonal divergence of moisture balances the meridional convergence of moisture just north of the Guinean coastline. In June, the strongest zonal divergence is located over the continental interior, but it does not completely compensate for

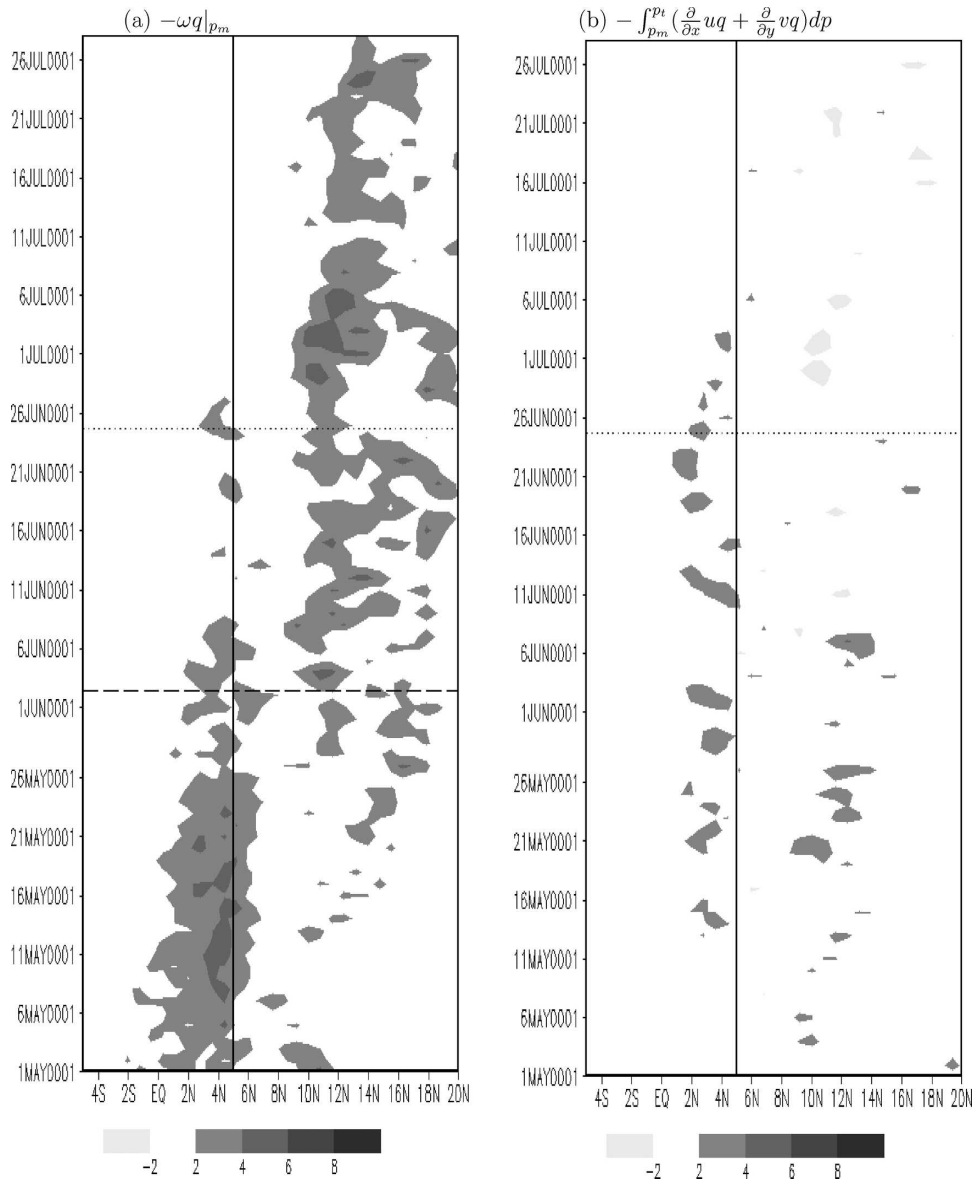


FIG. 6. (a) Vertical moisture flux into the upper layer and (b) and convergence of moisture at the upper layer (mm day^{-1}). All are averaged between 10°E and 10°W . The bold line represents the approximate latitude of the coastline, and the dashed lines represent the approximate dates of the appearance of vertical flux of moisture at 10°N and the jump in the precipitation maximum (see text).

the meridional convergence (Fig. 7a). Condensation in this layer also weakens the vertical moisture flux (Fig. 7d).

The net moisture convergence in the middle layer can be decomposed as follows:

$$-\int_{p_m}^{p_b} \left(\frac{\partial u q}{\partial x} + \frac{\partial v q}{\partial y} \right) dp = -\int_{p_m}^{p_b} q \left(\frac{\partial v}{\partial y} \right) dp - \int_{p_m}^{p_b} q \left(\frac{\partial u}{\partial x} \right) dp - \int_{p_m}^{p_b} v \left(\frac{\partial q}{\partial y} \right) dp - \int_{p_m}^{p_b} u \left(\frac{\partial q}{\partial x} \right) dp. \quad (4)$$

Figures 8a and 8b show the evolution of the first two terms on the rhs of Eq. (4). Taken together, these two

figures demonstrate that the meridional wind convergence accounts for almost all of the moisture conver-

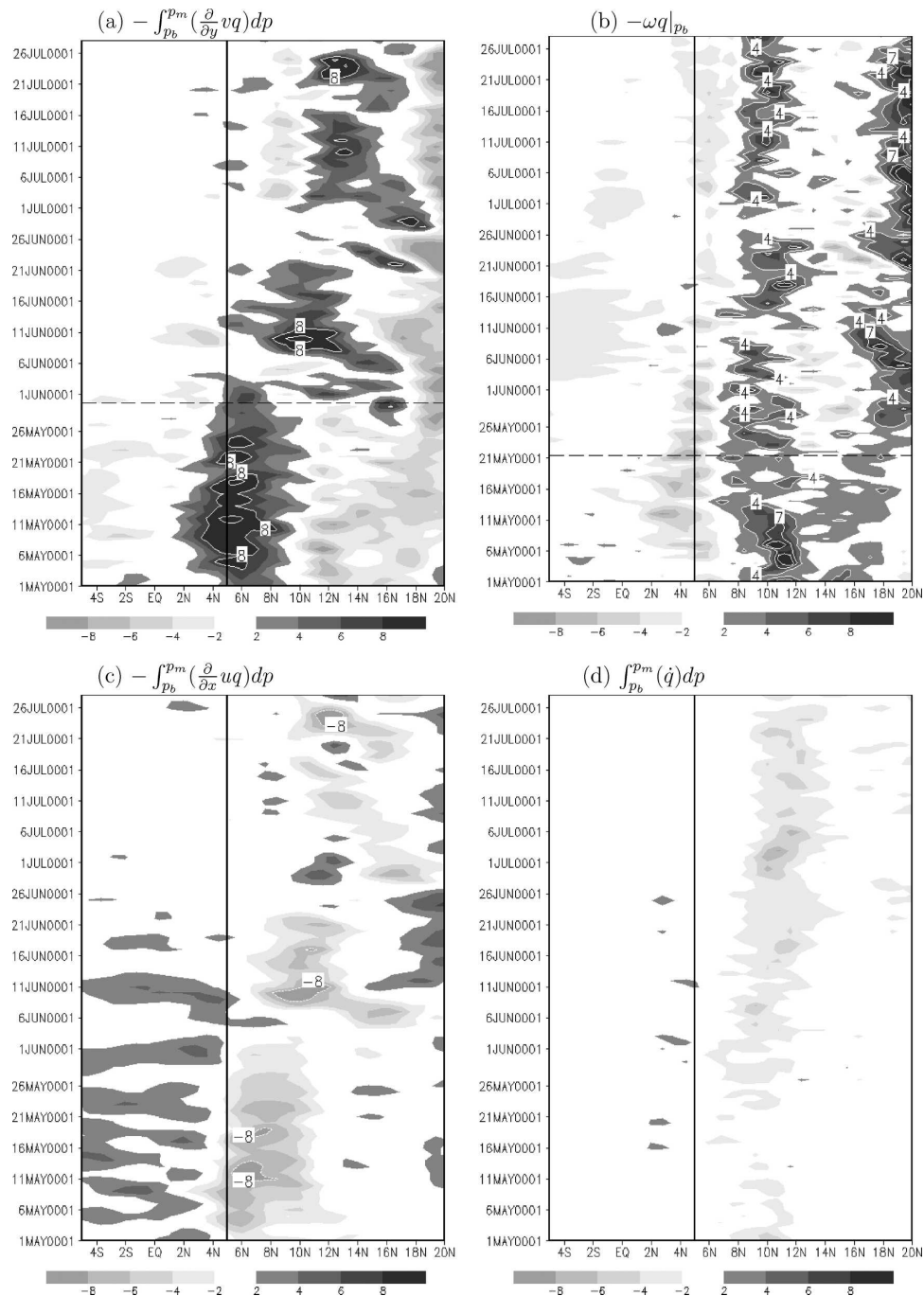


FIG. 7. Sources and sinks of moisture for the middle layer (mm day^{-1}). All are averaged between 10°E and 10°W . The bold line represents the approximate latitude of the coastline, and the dashed line represents the approximate date of the appearance of middle-layer moisture convergence at the continental interior.

gence while zonal wind divergence is a major sink north of the coastline (Fig. 8c). The third term in the rhs of Eq. (4), the meridional advection of moisture, is only important near the northern edge of the continental regime where gradients in moisture distribution are sig-

nificant, and the last term is negligibly small throughout (Fig. 8d).

Figures 9a and 9b show the meridional and zonal wind convergence. Comparison of Figs. 8a and 9a shows that evolution of the shift of the meridional mois-

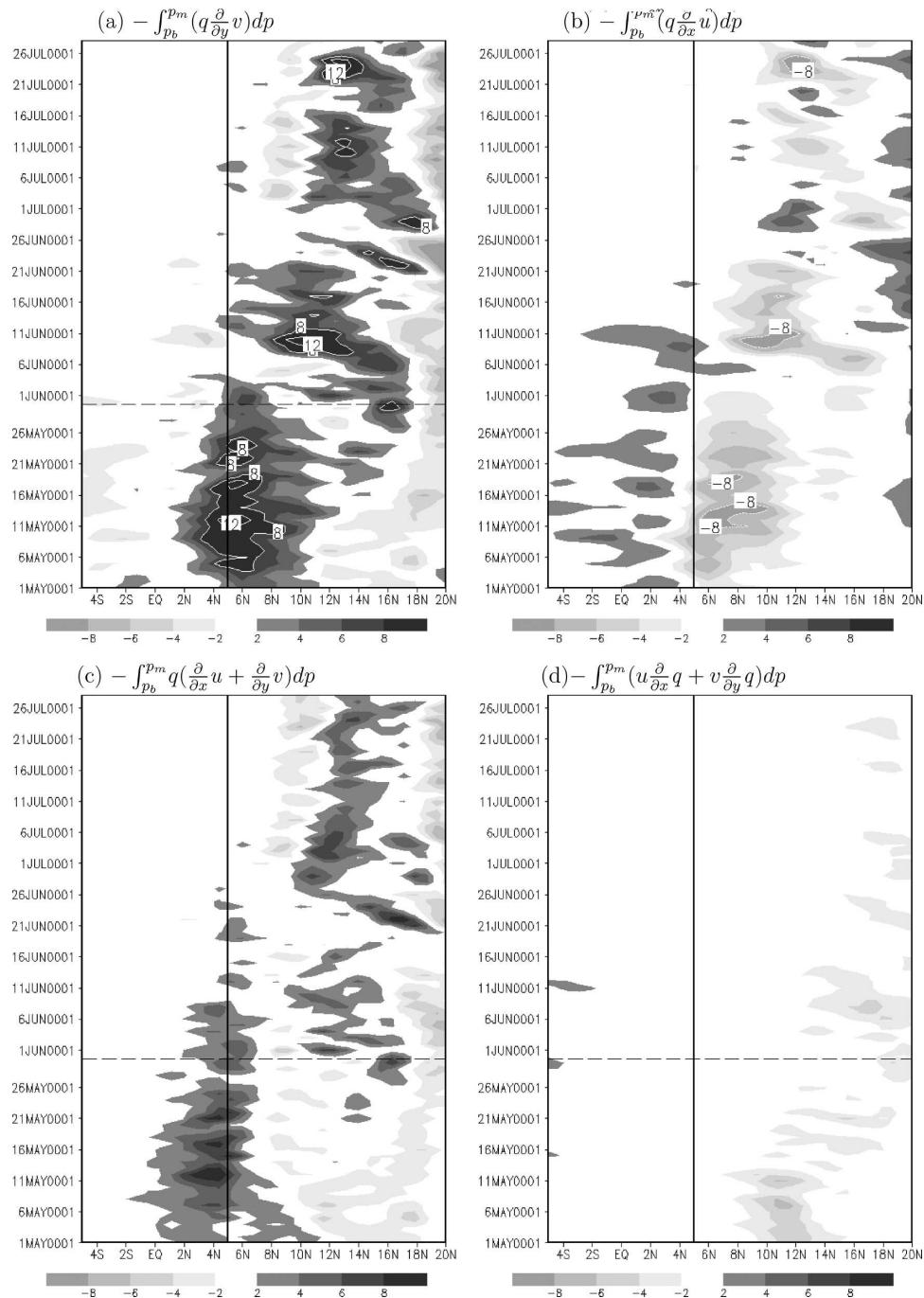


FIG. 8. Components of the moisture convergence in the middle layer (mm day^{-1}). All are averaged over 10°E and 10°W . The bold line represents the approximate latitude of the coastline and the dashed line represents the approximate date of the shift of middle-layer meridional moisture convergence into the continental interior.

ture convergence from the coast into the continent is tied to a similar shift in the meridional wind convergence. The zonal moisture divergence near 10°N is located at the southern edge of the African easterly jet and is related to the easterly acceleration of the flow

across West Africa, which results in a net loss of moisture (Fig. 9b). One important question is, then, why does the meridional convergence suddenly move into the continental interior near 30 May?

More general than Eq. (1), which assumes that the

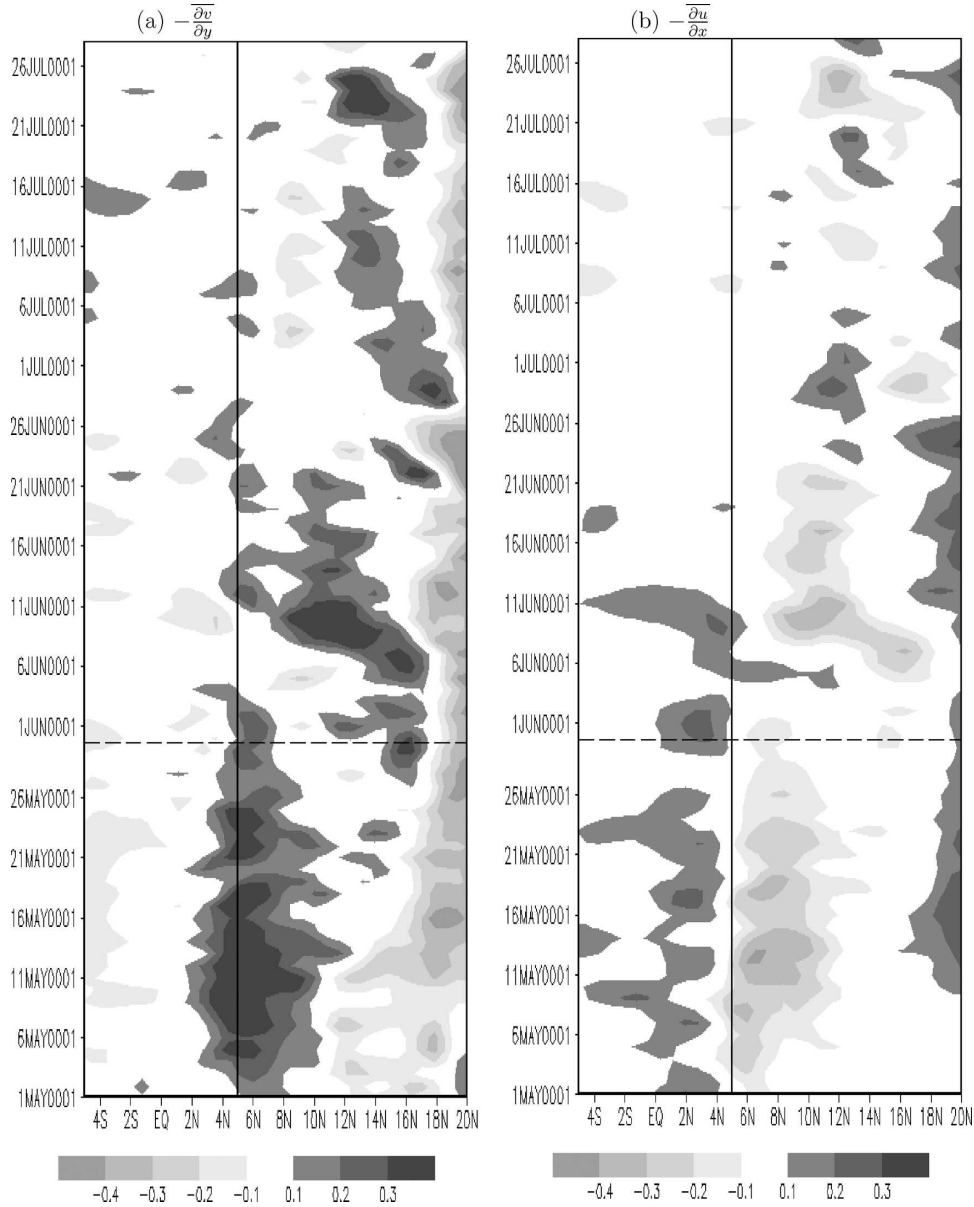


FIG. 9. Components of the wind convergence in the middle layer (day^{-1}). All are averaged over 10°E and 10°W . The bold line represents the approximate latitude of the coastline, and the dashed line represents the approximate date of the shift of middle-layer meridional wind convergence into the continental interior.

zonal flow is zonally uniform and geostrophic, perturbations are small, and friction, F_y , is negligible, the large-scale meridional flow is governed by the momentum equation

$$\frac{D\bar{v}}{Dt} = -f\bar{u} - \frac{\partial\bar{\phi}}{\partial y} + \bar{F}_y, \quad (5)$$

where the overbars indicate vertical averaging over the middle layer (825 to 525 hPa). The left-hand-side term

of Eq. (5) is displayed in Fig. 10a, from the RCM simulation. To eliminate most of the 3–5-day fluctuations, the plot is smoothed using a 10-point running mean. During the last days of May, the sign of the meridional acceleration of parcels along the coastline changes from positive (northward) to negative (southward).

For example, in line with the idea of inertial instability, consider a parcel of air located at point X on the zero contour of acceleration (Fig. 10a). Initially its ac-

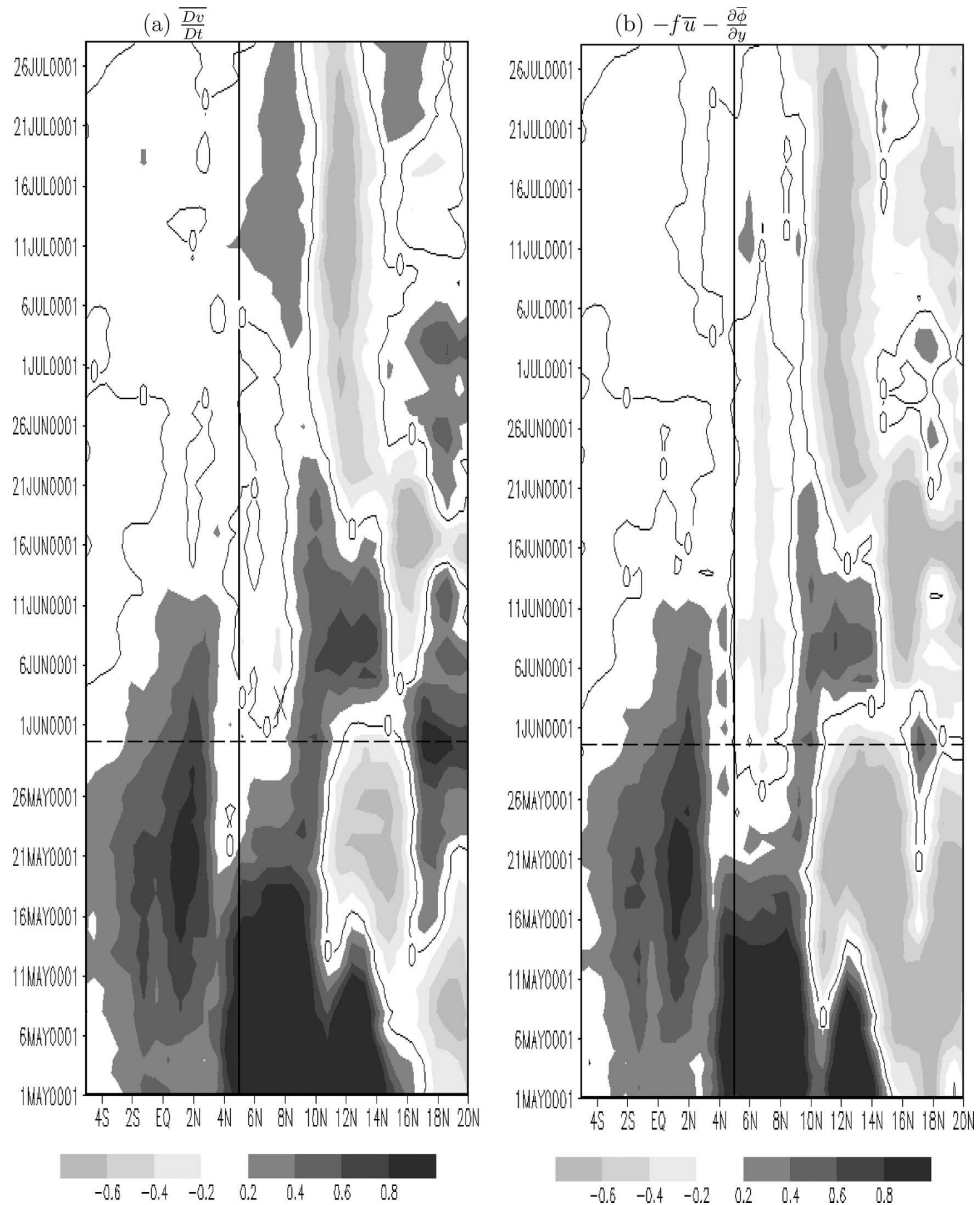


FIG. 10. (a) Acceleration and (b) force in units of 10^{-4} m s^{-2} . Both are averaged between 10°E and 10°W as well as 825 and 525 hPa. The bold line represents the approximate latitude of the coastline and the dashed line represents the approximate date of the shift of middle-layer meridional wind convergence into the continental interior.

celeration is zero. Any northward displacement would move the parcel into a region of positive net force and cause it to accelerate farther into the continent. Likewise, a parcel displaced southward is also accelerated farther southward. Therefore, because of inertial instability the coastal region (the region surrounded by the contour of zero acceleration) becomes unfavorable for meridional convergence in the end of May and the me-

ridional wind convergence jumps into the continental interior where convergence is sustainable.

Comparing Fig. 10b, which shows the sum of the first two right-hand-side terms of Eq. (5), with Fig. 10a indicates that the change in sign of the meridional acceleration is related to a change in the balance between the Coriolis and pressure gradient forces, while friction delays the process by about 3 days. Thus, the condition

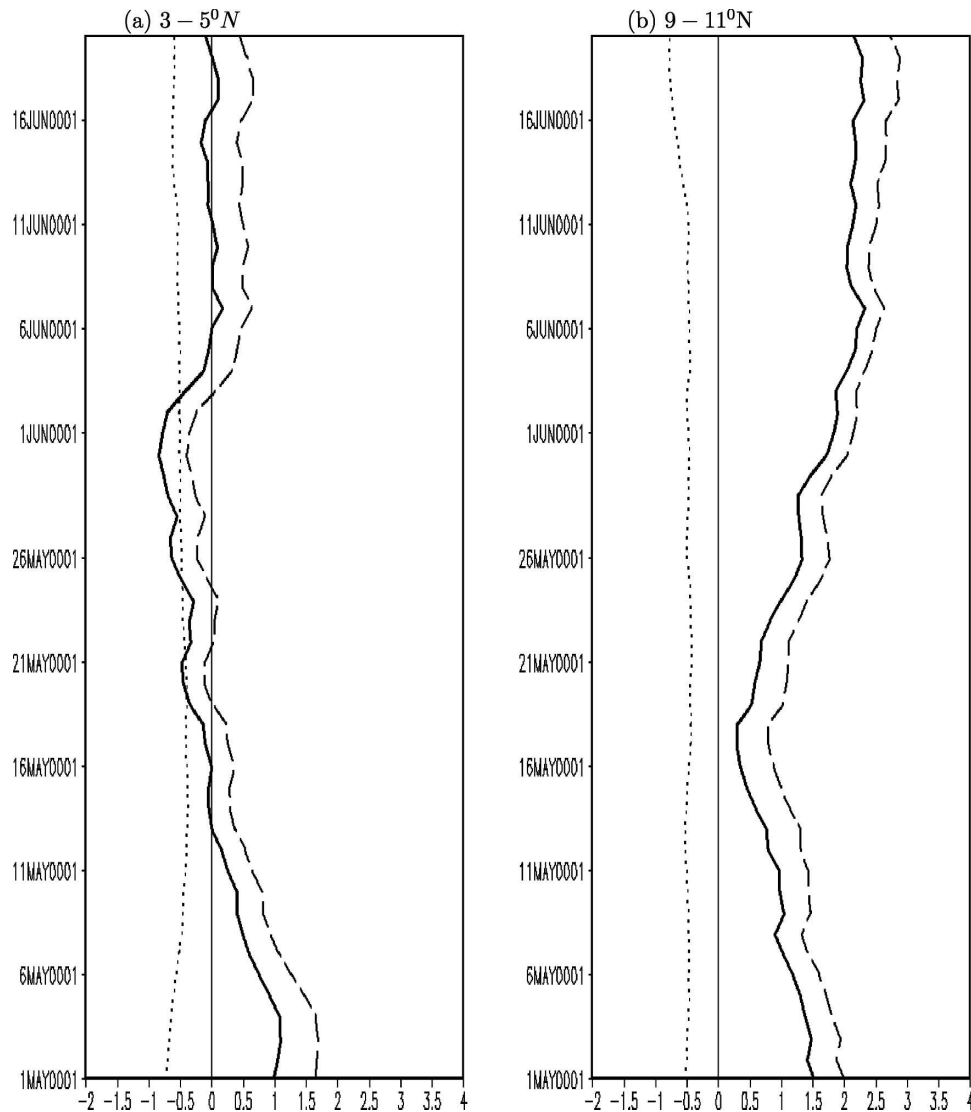


FIG. 11. Sources and sinks of mean potential temperature tendency θ (K day^{-1}) (a) along the coast (3° – 5°N) and (b) in the continental interior (9° – 11°N). The solid lines are total tendency; the dashed and dotted lines are condensational heating and radiative cooling, respectively. All are averaged over 10°E to 10°W .

for northward acceleration and the associated shift in meridional convergence is a change in sign of $-f\bar{u} - (\partial\bar{\phi}/\partial y)$. For a geostrophic, zonally uniform flow, this condition can be simplified to the change in sign of absolute vorticity discussed in the background section. The significant meridional acceleration over both the ocean and the continent throughout the period of simulation, however, makes the assumption of purely zonal flow during the premonsoon period questionable. The analysis is performed over a layer bounded by isobaric surfaces; therefore, the distribution of geopotential height, $\bar{\phi}$, is related to the mean potential temperature of the layer, $\bar{\theta}$. To identify the mechanisms that control

potential temperature over the continent and the ocean, the preonset thermodynamic balance over the middle layer is considered, given by

$$\int_{p_b}^{p_m} \frac{D\theta}{Dt} dp = \int_{p_b}^{p_m} \left[\frac{\theta Q_r}{c_p T} + \frac{\theta Q_c}{c_p T} + \frac{\theta Q_d}{c_p T} \right] dp. \quad (6)$$

The right-hand side of Eq. (6) consists of the potential temperature tendencies due to radiative, condensational, and diffusive heating, respectively. The condensational and radiative heating terms are plotted in Figs. 11a and 11b for the oceanic (3° – 5°N) and continental

regimes (9°–11°N) in the middle layer, respectively, along with the total potential temperature tendency. Diffusive heating is small and positive, as is expected for the free atmosphere. The values are smoothed by a 10-point running mean.

In both regions the supply of heat is mainly by middle-layer condensation, with generally constant radiative cooling reducing the overall potential temperature tendency. Throughout May, condensational heating rates over the oceanic regime decline (Fig. 11a) while those over the continent start to rise by the middle of May (Fig. 11b). The central questions then are: What are the physical processes that govern the condensation in the middle layer? What makes the oceanic and continental midtropospheric condensation

evolve in an opposite manner? These issues are discussed in the next subsection.

b. Premonsoon jump processes

In the last subsection it was shown that the most important source of the heating that fuels the inertial instability is condensation in the middle layer within the continent. As shown in Figs. 5b and 11b, this condensation starts increasing near the middle of May. In this section the mechanisms that control this increase in continental condensation through last two weeks of May are analyzed.

The moisture budget equation in the middle layer, Eq. (3), can be rewritten as

$$\int_{p_b}^{p_m} (\dot{q}) dp = \omega q_{p_b} + \int_{p_b}^{p_m} \left(\frac{\partial}{\partial x} uq \right) dp + \int_{p_b}^{p_m} \left(\frac{\partial}{\partial y} vq \right) dp - \omega q_{p_m} + \int_{p_b}^{p_m} \left(\frac{\partial}{\partial t} q \right) dp + \int_{p_b}^{p_m} F_q dp. \quad (7)$$

Continental condensation in May is controlled by the first three terms on the rhs, that is, moisture flux from the boundary layer (Fig. 7b) and divergence in the middle layer (Fig. 8c). Near the middle of the month, the gradually increasing moisture supply from the boundary layer becomes larger than the loss by divergence, and this leads to the onset of condensation in the middle layer on 20 May in the simulation. The other terms in Eq. (7) are relatively small.

In contrast, the roles of the boundary layer flux and moisture convergence in the middle layer are reversed along the coast. Here, the main source of moisture for

the oceanic middle layer is meridional convergence (Fig. 7a) and the net vertical moisture flux is directed downward into the boundary layer (Fig. 7b). The sources of the moisture that converges along the coast are the upward boundary layer fluxes to the south and, especially, to the north of the precipitation maximum along the coast. Therefore, the boundary layer supplies moisture to the continental middle layer and deprives the oceanic region. In this way, boundary layer fluxes facilitate the processes that lead to the monsoon jump.

The moisture budget equation for the boundary layer can be written as

$$-\omega q_{p_b} = - \int_{p_s}^{p_b} \left(\frac{\partial}{\partial x} uq \right) dp - \int_{p_s}^{p_b} \left(\frac{\partial}{\partial y} vq \right) dp - \omega q_{p_s} + \int_{p_s}^{p_b} (\dot{q}) dp - \int_{p_s}^{p_b} \left(\frac{\partial}{\partial t} q \right) dp - \int_{p_s}^{p_b} F_q dp, \quad (8)$$

where p_s is the pressure at the lowest model level. Over the ocean p_s is 995 hPa and over land it varies with topography. Figures 12a and 12b show the zonal and meridional moisture convergence terms, respectively. The main source of moisture for the flux from the boundary layer into the middle layer within the continent (~10°–12°N) is meridional moisture convergence (Fig. 12a). For the most part, the zonal flow diverges moisture in this layer (Fig. 12b). Fluxes from the surface also have a small positive contribution over the continental interior (not shown). The sum of the last three terms in Eq. (8) is important only where moisture gradients are relatively large (not shown).

The shallow meridional circulation that drives mois-

ture into the continent well before the monsoon jump has recently been observed by Zhang et al. (2006, manuscript submitted to *Quart. J. Roy. Meteor. Soc.*). Using sounding data, they demonstrated that this circulation exists throughout the year. This is consistent with our modeling result that the shallow meridional convergence at 10°N exists throughout the period of simulation without significant variations (Fig. 12a). They also observe that the monsoon onset is associated with the deepening of this otherwise boundary layer circulation. This is also consistent with our modeling result that during the monsoon season, the meridional moisture convergence extends well into the lower free troposphere (Fig. 8a).

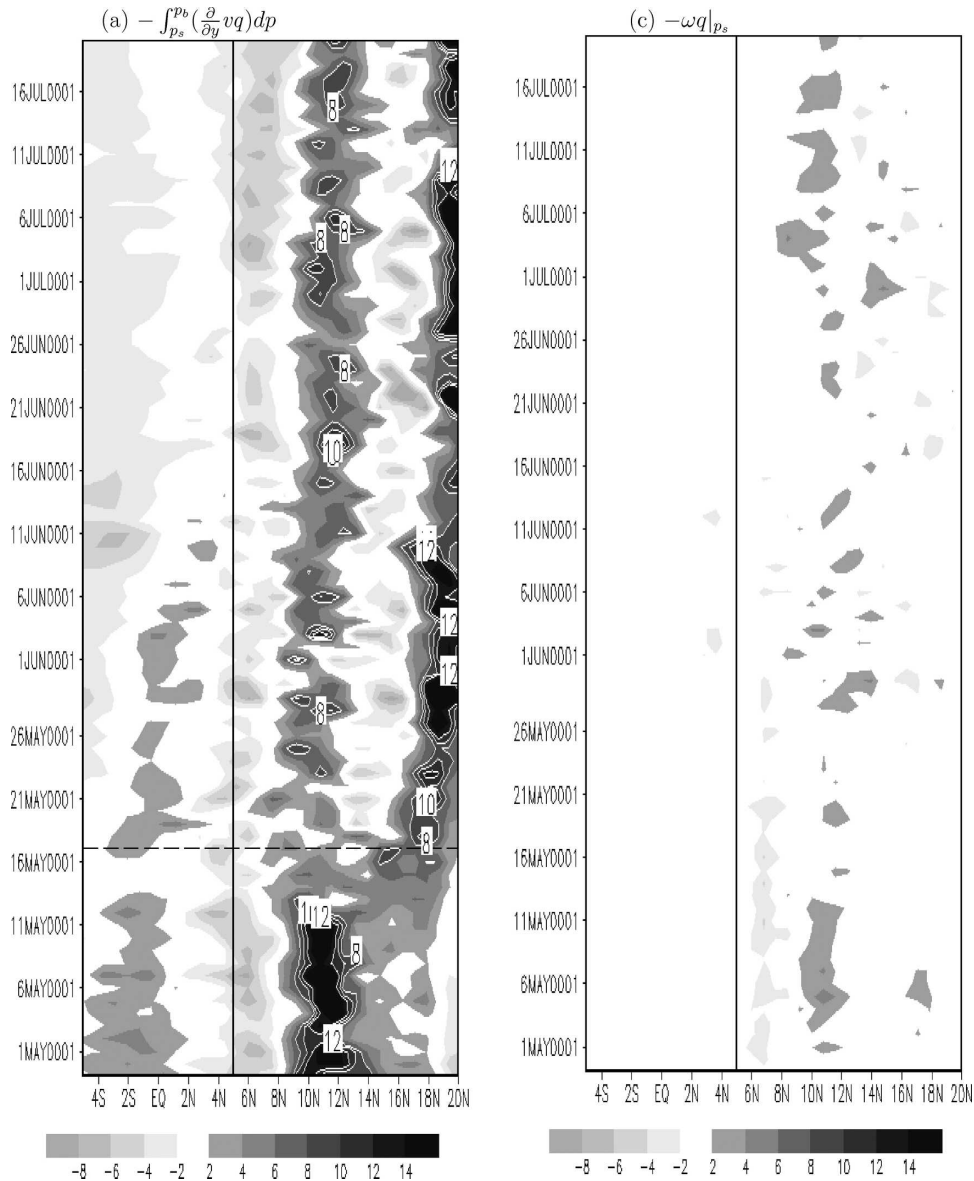


FIG. 12. Sources and sinks of moisture in the boundary layer (mm day^{-1}). All are averaged between 10°E and 10°W . Solid lines represent the approximate latitude of the coastline.

Having shown that the ultimate source of moisture (and heat) for the monsoon jump is meridional boundary layer convergence in the continental interior, the physical processes that control the evolution and the latitudinal location of this meridional convergence are investigated. The dynamics in the boundary layer is closely tied to the surface heat budget, which determines the amount of sensible heat transferred into the boundary layer. This supply of heat is balanced by the divergence of heat by the shallow circulation. For the time scales of interest, sensible (Q_{sh}) and latent (Q_{lh}) heat fluxes from the surface approximately balance the

sum of the net shortwave (Q_{sw}) and longwave (Q_{lw}) radiative heating supplied into the surface,

$$Q_{\text{sh}} + Q_{\text{lh}} \approx Q_{\text{sw}} + Q_{\text{lw}}. \quad (9)$$

Figure 13a shows the evolution of sensible heating, Fig. 13b the net radiative heating, Fig. 13c shortwave heating only, and Fig. 13d the latent heat lost by the surface due to evaporation. The sensible heating maximum remains in the continental interior (10°N) without significant latitudinal change (Fig. 13a). The location of the sensible heating maximum is determined by two factors; because of relatively low albedo the net shortwave

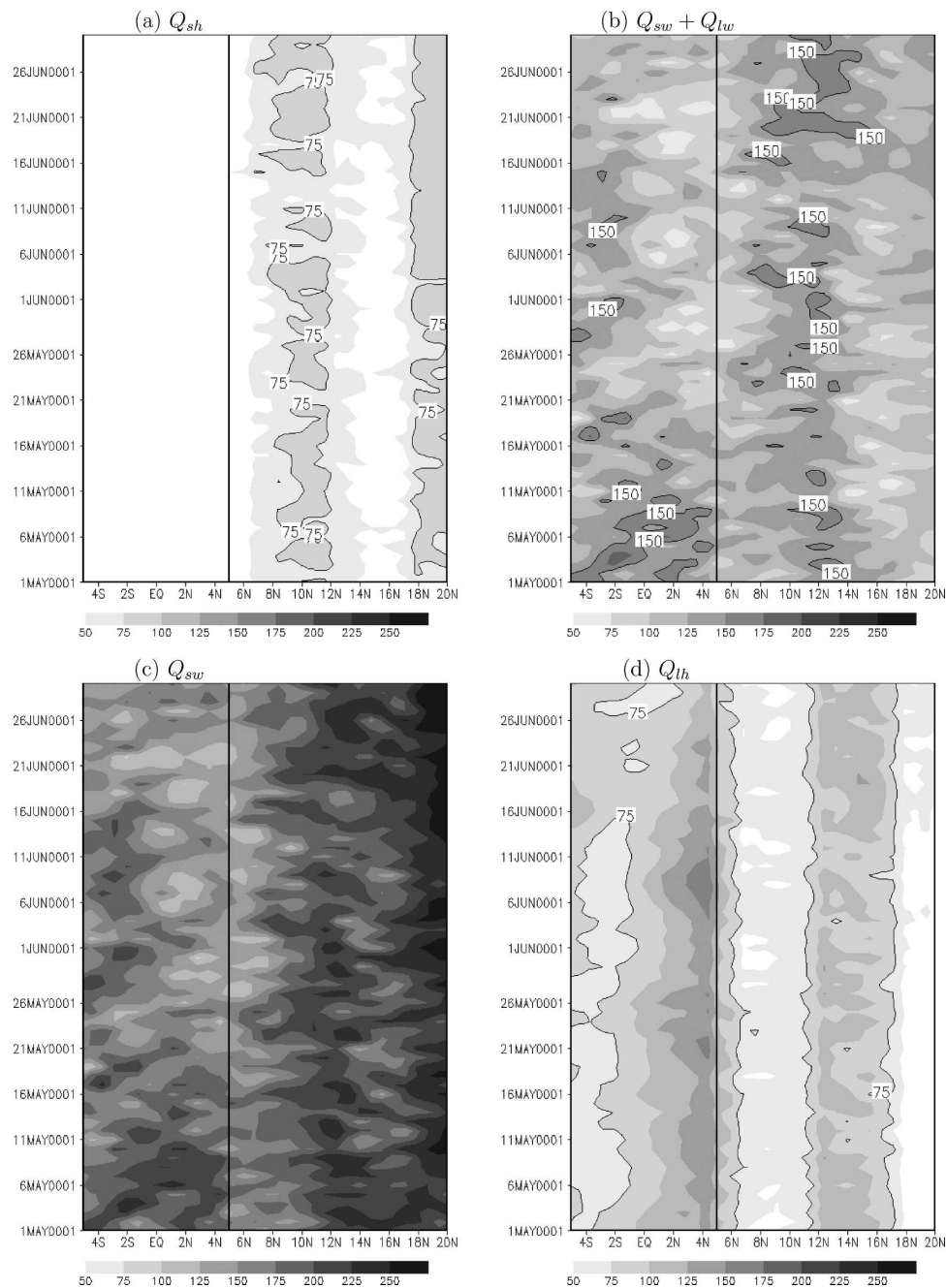


FIG. 13. (a) Surface upward sensible heating, (b) net downward radiative heating, (c) net downward shortwave heating, and (d) net upward latent heating fluxes in units of W m^{-2} . Solid lines represent the approximate latitude of the coastline.

radiation and the net total radiation have their maxima between 10° and 15°N (Figs. 13b,c). But about half of this radiation input into the surface is used in evaporation, which has its local maxima values between 12° and 18°N as well as south of 6°N (Fig. 13d). This introduces a sensible heating maximum between 6° and 12°N, which is also the location of the preonset shallow me-

ridional convergence, the subsequent deep monsoon meridional convergence, deep convection, and precipitation.

c. Summary of the process

The above discussion provides an analysis of the monsoon jump process. Based on this, one can con-

struct the following summary of the process. The dates are approximate.

- Because of the distribution of albedo and surface moisture availability, a sensible heating maximum persists over the Sahel. This sensible heating drives a shallow meridional circulation and moisture convergence at that latitude. The moisture is transported upward into the middle layer, where it diverges (Figs. 12a and 13a).
- During the second half of May, the increasing moisture flux from the boundary layer exceeds the divergence in the middle layer and results in a net supply of moisture and condensation (Figs. 5b and 7b). This condensation warms up the continental middle layer, while the evaporation of rain and radiation cool the middle layer along the coast (Fig. 11).
- The resulting pressure gradient results in an inertial instability, which abruptly shifts the meridional wind convergence maximum from the coast into the continental interior around 30 May. This introduces a net total moisture convergence, net upward moisture flux, and condensation in the upper layer, and the enhancement of precipitation in the continental interior (Figs. 10, 8, and 5a).
- During June, because of the shift of the meridional convergence into the continent and downward flux of moisture into the boundary layer, upper-layer condensation and precipitation along the coast gradually disappear.

5. Discussion and conclusions

The above modeling results agree to some extent with the previous observational analysis of Sultan and Janicot (2003), who also find that the West African monsoon jump involves the release of potential instability due to the supply of moisture by southerly surface winds followed by an inertial instability. In addition, our model agrees with their finding that surface heating drives the preonset moist southerly winds. However, in our model topography does not play a primary role in the process.

In their model simulations, Sijikumar et al. (2006) showed that the monsoon onset is characterized by a deepening of the heat low and accompanied by increased monsoon flow from the eastern Atlantic. Our results agree with that except in our model, the abrupt change is in the depth of the predominantly southerly monsoon flow from the Gulf of Guinea.

The above-discussed processes of the monsoon onset are not unique to West Africa. A similar onset mechanism has also been observed in the South Asian mon-

soon. The South Asian monsoon circulation is known to be driven by heating above the boundary layer, while the premonsoon shallow circulation provides the moisture that in turn introduces inertial instability in the free troposphere (see Webster et al. 1998, and references therein).

This work extends the previous theoretical studies that suggest boundary layer entropy gradients are responsible for the onset of the monsoon. The change in the sign of $-\bar{f}\bar{u} - (\partial\bar{\phi}/\partial y)$ shown here to be the primary condition for the inertial instability reduces to the negative absolute vorticity condition that implies the breakdown of a strictly zonal geostrophic flow in the idealized systems described by Emanuel (1995) and Eltahir and Gong (1996). In this case, however, the preonset system is not necessarily strictly zonal or geostrophic. In fact, the inertial instability owes its appearance to the ageostrophy of the system. By divesting moisture, zonal wind divergence plays a delaying role in the onset process. In our study, the premonsoon shallow meridional circulation, which is observed in atmospheric sounding study by Zhang et al. (2006, manuscript submitted to *Quart. J. Roy. Meteor. Soc.*), plays a very important role in transporting moisture from the ocean into the continent weeks before the onset of deep convection over the continent. Considering the contribution of the simple theoretical models to the improvement of prediction (Fontaine et al. 1999), one can hope to achieve even better skill by considering the above-discussed interactions among the boundary layer and the various layers of the free troposphere in future work on prediction.

In this study, the seasonal evolution of SST is prescribed. As noted in the background section, there are some modeling and observational results that suggest the monsoon has significant influence on SSTs in the Gulf of Guinea (e.g., Okumura and Xie 2004; Mitchell and Wallace 1992). This interaction should be further investigated using coupled models.

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